

state s and
scenario ω

check preconditions
of all 540 actions:
45 techniques $\times$ 12 hosts

mask $\mu(s, \omega)$:
one bit per action

masked logits
set to $-\infty$

legal action
sampled

computed by the environment at every step; never learned

An action whose preconditions do not hold

under the prerequisite mask
receives no probability mass
and cannot be proposed at all

0.0% of actions infeasible

under reward shaping
is proposed, executed, refused,
and penalised afterwards

10.3% infeasible after training, from 99%